

CSC311H5 Introduction to Machine Learning

Lecture 9

University of Toronto Mississauga

Test #2 Reminders

- Friday Mar 20, 7:15-8:45pm, IB110/IB120
- Covers materials up to the end of week 8, with a focus on weeks 5-8
- See Quercus for the list of learning objectives tested
- Make sure to **bring your own non-programmable calculator**
 - ▶ calculators cannot be shared
- You may bring one single-sided cheat sheet

Cheat Sheet Rules

- Must be one single-sided sheet of paper (or no more than 4000 characters if you cannot use a sheet of paper for accessibility reasons).
- Must be created by and unique to the student who is using it. You may not copy another student's cheat sheet.
- Must have student's name and student number clearly at the top.
- Can be hand-written or typed. Whatever fits on one page is fine.
- Will be collected with the test. If you want to keep a copy, you must make your own copy in advance.

Section 1

Mistakes in ML: Algorithmic Fairness and More

Machine Learning Impact

What issues come to mind when thinking about how machine learning models positively impact society?

Machine Learning Impact

What issues come to mind when thinking about how machine learning models negatively impact society, either through intended use, misuse, or unintended consequences?

Mistakes in Machine Learning

We will focus on “mistakes” or unintended consequences in this lecture.

Though issues stemming from intended use and misuse are also important!

Mistakes that we focus on:

- Issues related to overestimating model performance
- Issues related to bias/fairness

These mistakes are related to:

- Unrepresentative training data
- Improper/incomplete testing (e.g., data leakage)
- Over-reliance on accuracy measures (e.g., leading to issues of *alignment*)

Example: Football vs Bald Head

TECH / ARTIFICIAL INTELLIGENCE

AI camera operator repeatedly confuses bald head for soccer ball during live stream



Where's the ball? There's the ball, get it! GIF: [YouTube](#) / [Chuck1ehands](#)

/ Like a distracted AI with a crush

By [James Vincent](#), a senior reporter who has covered AI, robotics, and more for eight years at The Verge.

Nov 3, 2020, 8:07 AM EST



0 Comments (0 New)

<https://www.theverge.com/tldr/2020/11/3/21547392/ai-camera-operator-football-bald-head-soccer-mistakes>

Example: Bias in Image Recognition

The New York Times

Artificial Intelligence > | A.I. Faces Quiz | How the A.I. Race Began | Key Figures in the Field | One Year of ChatGPT

Google's Photo App Still Can't Find Gorillas. And Neither Can Apple's.



Desiree Rios/The New York Times

Eight years after a controversy over Black people being mislabeled as gorillas by image analysis software — and despite big advances in computer vision — tech giants still fear repeating the mistake.

By Nico Grant and Kashmir Hill

May 22, 2023

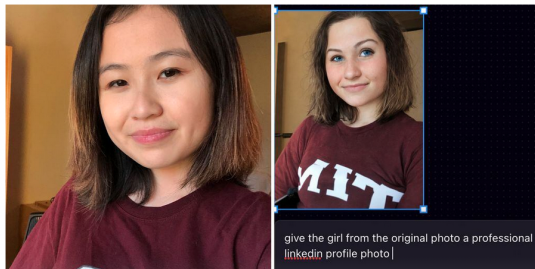
<https://www.nytimes.com/2023/05/22/technology/ai-photo-labels-google-apple.html>

Example: Generative AI

An MIT student asked AI to make her headshot more 'professional.' It gave her lighter skin and blue eyes.

Computer science major Rona Wang said her experience shows how the emerging technology can have racial blind spots.

By **Spencer Suen** | Globe Staff, Updated July 19, 2023, 10:23 a.m.



MIT student Rona Wang asked an AI image creator app called Playground AI to make a photo of her look "professional." It gave her paler skin and blue eyes, and "made me look Caucasian." RONA WANG

<https://www.bostonglobe.com/2023/07/19/business/an-mit-student-asked-ai-make-her-headshot-more-professional-it-gave-her-lighter-skin-blue-eyes/>

Example: Data Leakage

Field	Paper	Number of papers reviewed	Number of papers with pitfalls	(L1.1) No test set	(L1.2) Pre-proc. on train-test	(L1.3) Feature set. on train-test	(L1.4) Duplicates	(L2) Illegitimate features	(L2.1) Temporal leakage	(L2.2) Non-ind. b/w train-test	Comput. reproducibility issues	Data quality issues	Metric choice issues	Standard dataset used?
Medicine	Bouwmeester et al. (2012)	71	27	o							o			
Neuroimaging	Whelan & Garavan (2014)	–	14	o	o									
Autism Diagnostics	Bone et al. (2015)	–	3			o			o		o	o	o	o
Bioinformatics	Blagus & Lusa (2015)	–	6		o									
Nutrition Research	Ivanescu et al. (2016)	–	4	o							o	o		
Software Eng.	Tu et al. (2018)	58	11				o				o	o	o	o
Toxicology	Alves et al. (2019)	–	1			o					o	o		
Satellite Imaging	Nalepa et al. (2019)	17	17					o			o		o	
Tractography	Poulin et al. (2019)	4	2	o							o	o	o	o
Clinical Epidem.	Christodoulou et al. (2019)	71	48		o						o			
Brain-computer Int.	Nakanishi et al. (2020)	–	1	o									o	
Histopathology	Oner et al. (2020)	–	1					o						
Neuropsychiatry	Poldrack et al. (2020)	100	53	o	o						o	o	o	o
Medicine	Vandewiele et al. (2021)	24	21		o			o	o	o	o	o	o	o
Radiology	Roberts et al. (2021)	62	62	o		o			o	o			o	
IT Operations	Lyu et al. (2021)	9	3				o						o	
Medicine	Filho et al. (2021)	–	1			o								
Neuropsychiatry	Shim et al. (2021)	–	1		o					o				
Genomics	Barnett et al. (2022)	41	23		o						o			
Computer Security	Arp et al. (2022)	30	30	o	o	o	o	o	o	o	o	o	o	o

Table 1. Survey of 20 papers that identify pitfalls in the adoption of ML methods across 17 fields, collectively affecting 329 papers. In each field, papers adopting ML methods suffer from data leakage. The column headings for types of data leakage, shown in bold, are based on our taxonomy of data leakage. We also highlight other issues that are reported in the papers, including issues with computational reproducibility (the availability of code, data, and computing environment to reproduce the exact results reported in the paper), data quality (for example, small size or large amounts of missing data), metric choice (using incorrect metrics for the task at hand, for example, using accuracy for measuring model performance in the presence of heavy class imbalance), and standard dataset use, where issues are found despite the use of standard datasets in a field.

<https://arxiv.org/pdf/2207.07048.pdf>

Overestimating Model Performance

Is your test set and/or your testing strategy *really* representative of how your model will be used in practice?

Diversity: Is your test set diverse? Often it is difficult to think of all possible dimensions of “diversity”, but having a diverse team of model implementors help!

Data Leakage: Is your test data correlated to your training/validation data in unintuitive ways?

- In lab 3 (temperature prediction), why did we not randomly sample our test set?
- What is wrong if we build a stock market predictor using data from all stocks on the TSX?
- In the final project, we will have 3 data points generated per student; would be okay to randomly split the data points into train/validation/test set? What could be the problem?

Bias in Machine Learning Models

What do you think causes bias in ML models like this one?

TECH / APPLE / POLICY

Apple's credit card is being investigated for discriminating against women



Image: Apple

/ Customers say the card offers less credit to women than men

By [James Vincent](#), a senior reporter who has covered AI, robotics, and more for eight years at The Verge.

Nov 11, 2019, 5:57 AM EST



0 Comments (0 New)

<https://www.theverge.com/2019/11/11/20958953/apple-credit-card-gender-discrimination-algorithms-black-box-investigation>

Bias in Machine Learning Models

Suppose we remove a person's gender as a feature from the model. Can we guarantee that the model would no longer be biased with respect to gender?

Bias in Machine Learning Models

Suppose we remove a person's gender as a feature from the model. Can we guarantee that the model would no longer be biased with respect to gender?

No!

There may be other features that correlates with gender, or can otherwise predict a person's gender.

Bias in Machine Learning Models

Suppose we remove a person's gender as a feature from the model. Can we guarantee that the model would no longer be biased with respect to gender?

No!

There may be other features that correlates with gender, or can otherwise predict a person's gender.

So how can we measure bias? What does it even mean for a machine learning model to be unfair?

Section 2

Value Cards Exercise

Activity: ML for Recidivism Prediction

Activity Premise:

- Canadian Courts would like to explore using an AI model to help classify whether a defendant should be granted parole, based on their likelihood to reoffend.
- **Recidivism**: the act of a convicted criminal to commit another crime.
- The model's prediction will be used to aid a judge in making a decision, so the ultimate decision is still up to a judge.
- We are considering which of 8 possible models to use. Each model has a different performance characteristic.

Understanding Societal Values in AI

- There is no single definition of societal values that will apply equally well to different applications of AI/ML.
- Prioritizing one value in AI/ML systems often means making tradeoffs on competing values. It is therefore important to be explicit and transparent about priorities and assumptions.
- There are seldom clear-cut answers. It is therefore important to document your processes and considerations (including priorities and tradeoffs).

Identifying Stakeholders

- **Who is at risk of experiencing impacts:** considering both the people who will use the system and the people who will be directly or indirectly affected by the system, either by choice or not.
- **People often belong to overlapping groups:** different combinations of race, gender, and age, for example—and specific intersectional groups may be at greatest risk of experiencing different types of harm.
- **What are the tradeoffs** between expected benefits and potential harms for identified stakeholder groups?

Analyzing impacts

- What are the types of impact (e.g., allocation, quality of service, stereotyping, denigration, over- or underrepresentation) on different stakeholders
- Rate the degree of impact [no discernable - minor - moderate - major]
- Estimate the scale of impact [small - medium - large]
- Estimate the overall direction of impact [positive - mostly positive - mostly negative - negative]

Value Cards Activity

Form groups of 4. In your group, choose who will be group member:

- A
- B
- C
- D

Value Cards Activity: Part 1

Each of group members A, B, C, D will be in a different part of the classroom!

- Each group (A, B, C, D) will be given one of four *Personas*
- Read the personas together.
- Read the model cards together.
- Formulate an argument, from your persona's perspective, the advantages and disadvantage of each model.
- You will be bringing these argument to a negotiation session with the other personas in your original groups.

Value Cards Activity: Part 2

Go back to your original group.

Your goal is to see if you can **come to a consensus on which model should be deployed**

- Each group should have members from each persona.
- LISTEN: Each persona should have 1-2 minutes to discuss their values, concerns, and perspectives.
- NEGOTIATE: See if you can come to a consensus on which model should be deployed.

Remember: Your goal is not to “win”, but to:

- express your perspective, and be a strong advocate
- while also actively listening to your other group members

Value Cards Activity Debrief

- Did your group reach a consensus?
- If so, what model did you choose?

Section 3

Algorithmic Fairness

Ways to think about fairness

Disparate Treatment vs Disparate Impact

- Similar to the difference between “equality” and “equity”
- **Disparate Treatment:** are decisions correlated with subject’s sensitive attribute?
 - ▶ Is it fair to ask for a higher credit score from one group, because the group historically has higher default rates?
- **Disparate Impact:** are decisions disproportionately hurting people with sensitive attributes?
 - ▶ Is it fair to ask for the same credit score from all groups, if one group has been historically denied the same resources to build their credit scores?

It’s possible to treat people the same, but still have the decision impact groups in different ways.

Ways of measuring fairness

There is no consensus on how to measure fairness of a model.

Different measure of fairness can contradict each other!

We'll introduce three metrics today:

- Demographic Parity
- Accuracy Parity
- Individual Fairness

Fairness as Demographic Parity

Demographic Parity:

- Acceptance **rates** of applications from both groups must be equal
- Also known as “independence” (terminology from statistics)

Problem:

- Fairness is measured at a *group* level
- Model can hire qualified people from one group, and random people from the other

Fairness as Accuracy Parity

Accuracy Parity:

- Model should be **equally accurate** across both groups

Problem:

- False positives and false negatives have different impacts
- Does not help to close the gap between the two groups

Fairness as Individual Fairness

Individual Fairness:

- Similar individuals from different groups should be treated similarly

Problem:

- Hard to determine appropriate measure of “similarity” of inputs

Trade off

- There are more definitions of fairness than these (e.g. Equal Opportunity, Equalized Odds)
- The various definitions are inconsistent with each other
- Optimizing one definition of fairness means trading off accuracy and other notions of fairness
- But this doesn't mean we shouldn't try! Rather, we should analyze what we most value in a particular application and choose a notion of fairness that aligns well.

Ideas for more fair models

Pre-processing: remove information correlated to sensitive attributes

Add regularization term: add a “fairness” regularizer

- We saw regularizers that encourage smaller weights
- This kind of regularizer would encourage “fairness”

Post-processing: change the way we use a model to make predictions

Section 4

Test #2 Review

Neural Networks and Backprop

What is the relationship between these terms and ideas?

- Gradient Descent
- Backpropagation
- Computation Graph
- The Chain Rule

V6-T8

Consider the computation below, where each of $\mathbf{d}$, $\mathbf{e}$, $\mathbf{f}$, are column vectors, and $\mathbf{C}$ is a matrix. We will be computing the gradient $\nabla_{\mathbf{d}}\mathcal{L}$ step by step, via backpropagation.

$$\mathbf{e} = \mathbf{C}\mathbf{d}$$

$$\mathbf{f} = \sigma(\mathbf{e})$$

$$\mathcal{L} = \mathbf{f}^T(\mathbf{f} + \mathbf{e})$$

We will be applying the chain rule.

Remember the three steps to writing down vectorized quantities: (1) compute scalar partial derivatives, (2) write the result in a vector, (3) find patterns.

V6-T8(a) - Computing $\nabla_{\mathbf{f}} \mathcal{L}$

Start by writing down the (scalar) relationship between f_i and $\mathcal{L}$

$$\mathcal{L} = f_1(f_1 + e_1) + f_2(f_2 + e_2) + \cdots + f_D(f_D + e_D)$$

V6-T8(a) - Computing $\nabla_{\mathbf{f}}\mathcal{L}$

Start by writing down the (scalar) relationship between f_i and $\mathcal{L}$

$$\mathcal{L} = f_1(f_1 + e_1) + f_2(f_2 + e_2) + \cdots + f_D(f_D + e_D)$$

Now, take derivatives:

$$\frac{\partial \mathcal{L}}{\partial f_i} = 2f_i + e_i$$

Finally, vectorize:

$$\nabla_{\mathbf{f}}\mathcal{L} = \begin{bmatrix} 2f_1 + e_1 \\ 2f_2 + e_2 \\ \vdots \\ 2f_D + e_D \end{bmatrix} = 2\mathbf{f} + \mathbf{e}$$

V6-T8(b) - Computing $\frac{\partial f_i}{\partial e_j}$

Since $\mathbf{f} = \sigma(\mathbf{e})$ is applied element-wise, $f_i = \sigma(e_i)$

So for $i \neq j$ we have $\frac{\partial f_i}{\partial e_j} = 0$.

And for $i = j$ we have ...

V6-T8(b) - Computing $\frac{\partial f_i}{\partial e_j}$

Since $\mathbf{f} = \sigma(\mathbf{e})$ is applied element-wise, $f_i = \sigma(e_i)$

So for $i \neq j$ we have $\frac{\partial f_i}{\partial e_j} = 0$.

And for $i = j$ we have ...

$$\frac{\partial f_i}{\partial e_i} = \sigma(e_i)(1 - \sigma(e_i)) = f_i(1 - f_i).$$

V6-T8(c) - Computing $\nabla_{\mathbf{e}} \mathcal{L}$

Each e_j affects the loss through two paths: directly affecting $\mathcal{L}$ and by affecting $\mathbf{f}$, which affects $\mathcal{L}$. Thus,

V6-T8(c) - Computing $\nabla_{\mathbf{e}} \mathcal{L}$

Each e_j affects the loss through two paths: directly affecting $\mathcal{L}$ and by affecting $\mathbf{f}$, which affects $\mathcal{L}$. Thus,

$$\frac{\partial \mathcal{L}}{\partial e_j} = f_j + \frac{\partial \mathcal{L}}{\partial f_j} \frac{\partial e_j}{\partial f_j}$$

This above part used the fact that e_j directly affects $\mathcal{L}$ (through the first term), and that e_j only affects f_j (and not any of the other f_i s where $i \neq j$). Thus, we have

$$\frac{\partial \mathcal{L}}{\partial e_j} = f_j + (2f_j + e_j)f_i(1 - f_i)$$

Vectorizing, we get...

V6-T8(c) - Computing $\nabla_{\mathbf{e}} \mathcal{L}$

Each e_j affects the loss through two paths: directly affecting $\mathcal{L}$ and by affecting $\mathbf{f}$, which affects $\mathcal{L}$. Thus,

$$\frac{\partial \mathcal{L}}{\partial e_j} = f_j + \frac{\partial \mathcal{L}}{\partial f_j} \frac{\partial e_j}{\partial f_j}$$

This above part used the fact that e_j directly affects $\mathcal{L}$ (through the first term), and that e_j only affects f_j (and not any of the other f_i s where $i \neq j$). Thus, we have

$$\frac{\partial \mathcal{L}}{\partial e_j} = f_j + (2f_j + e_j)f_i(1 - f_i)$$

Vectorizing, we get...

$$\nabla_{\mathbf{e}} \mathcal{L} = \mathbf{f} + (2\mathbf{f} + \mathbf{e}) \odot \mathbf{f} \odot (1 - \mathbf{f})$$

Where $\odot$ is element-wise multiplication. (On tests, we will often ask you to write this using the *diag* function.)

V6-T8(d) - Computing $\frac{\partial e_j}{\partial d_k}$

Again, start by writing down the (scalar) relationship between f_i and $\mathcal{L}$

$$\mathbf{e} = \mathbf{C}\mathbf{d}$$

So each e_j is computed as follows:

V6-T8(d) - Computing $\frac{\partial e_j}{\partial d_k}$

Again, start by writing down the (scalar) relationship between f_i and $\mathcal{L}$

$$\mathbf{e} = \mathbf{C}\mathbf{d}$$

So each e_j is computed as follows:

$$e_j = \sum_k C_{jk} d_k$$

So we have ...

V6-T8(d) - Computing $\frac{\partial e_j}{\partial d_k}$

Again, start by writing down the (scalar) relationship between f_i and $\mathcal{L}$

$$\mathbf{e} = \mathbf{C}\mathbf{d}$$

So each e_j is computed as follows:

$$e_j = \sum_k C_{jk} d_k$$

So we have ...

$$\frac{\partial e_j}{\partial d_k} = C_{jk}$$

V6-T8(e) - Computing $\nabla_{\mathbf{d}}\mathcal{L}$

Applying the chain rule to compute $\frac{\partial \mathcal{L}}{\partial d_k}$:

$$\frac{\partial \mathcal{L}}{\partial d_k} = \sum_j \frac{\partial \mathcal{L}}{\partial e_j} \frac{\partial e_j}{\partial d_k} = \sum_j \frac{\partial \mathcal{L}}{\partial e_j} c_{jk}$$

This is an **inner product**, between $\nabla_{\mathbf{e}}\mathcal{L}$ and the k – *th* column of $\mathbf{C}$.

Let's define $\mathbf{C}_{\cdot,k}$ to be the k th column of $\mathbf{C}$, so we can write ...

V6-T8(e) - Computing $\nabla_{\mathbf{d}}\mathcal{L}$

Applying the chain rule to compute $\frac{\partial \mathcal{L}}{\partial d_k}$:

$$\frac{\partial \mathcal{L}}{\partial d_k} = \sum_j \frac{\partial \mathcal{L}}{\partial e_j} \frac{\partial e_j}{\partial d_k} = \sum_j \frac{\partial \mathcal{L}}{\partial e_j} c_{jk}$$

This is an **inner product**, between $\nabla_{\mathbf{e}}\mathcal{L}$ and the k – *th* column of $\mathbf{C}$.

Let's define $\mathbf{C}_{\cdot,k}$ to be the k th column of $\mathbf{C}$, so we can write ...

$$\nabla_{\mathbf{d}}\mathcal{L} = \begin{bmatrix} \nabla_{\mathbf{e}}\mathcal{L} \cdot \mathbf{C}_{\cdot,1} \\ \nabla_{\mathbf{e}}\mathcal{L} \cdot \mathbf{C}_{\cdot,2} \\ \dots \end{bmatrix}$$

This is a matrix-vector multiplication

$$\nabla_{\mathbf{d}}\mathcal{L} = \mathbf{C}^T \nabla_{\mathbf{e}}\mathcal{L}$$

Probabilistic Modelling

- Why might we model $P(X|Y)$ or $P(\mathbf{x}|c)$ instead of $P(c|\mathbf{x})$?
 - ▶ Note: the first notation is more reminiscent of what you have seen in a statistics/probability course. The second notation is more consistent with what we see in ML applications.
- What does it mean when we say that a model is *generative*? What does this word have to do with *generating data* and the data distribution p_{data} from the bias-variance decomposition discussion?
- What is the conditional independence assumption?

W7-V1 and W7-V2

Suppose that you have the following data set:

X_1 (ChestPain)	X_2 (Smoking)	Y (Heart Disease)	Count
1	1	0	10
1	1	1	250
1	0	0	40
1	0	1	50
0	1	0	30
0	1	1	200
0	0	0	5
0	0	1	415

- W7-V1 models $P(Y|X_1, X_2)$ directly.
- W7-V2 models $P(X_1, X_2|Y = 0)$ and $P(X_1, X_2|Y = 1)$.

In this case, there is not much difference between these two models.

W7-V1(a)

Suppose we wish to model $P(Y|X_1, X_2)$ directly, as we did in class. Thus, each $P(Y|X_1 = 1, X_2 = 1)$, $P(Y|X_1 = 1, X_2 = 0)$, $\dots$, will be a separate distribution to be learned. List all the parameters in this model that will need to be learned.

W7-V1(a)

Suppose we wish to model $P(Y|X_1, X_2)$ directly, as we did in class. Thus, each $P(Y|X_1 = 1, X_2 = 1)$, $P(Y|X_1 = 1, X_2 = 0)$, $\dots$, will be a separate distribution to be learned. List all the parameters in this model that will need to be learned.

So, we will need four Bernoulli parameters

$$P(Y = 1|X_1 = 0, X_2 = 0) = \theta_{00}$$

$$P(Y = 1|X_1 = 0, X_2 = 1) = \theta_{01}$$

$$P(Y = 1|X_1 = 1, X_2 = 0) = \theta_{10}$$

$$P(Y = 1|X_1 = 1, X_2 = 1) = \theta_{11}$$

W7-V1(b)

Suppose we wish also wish to compute/model $P(Y)$ directly. Explain why this can be done by modeling $P(X_1, X_2)$. How many parameters are required in this distribution?

W7-V1(b)

Suppose we wish also wish to compute/model $P(Y)$ directly. Explain why this can be done by modeling $P(X_1, X_2)$. How many parameters are required in this distribution?

We will need three parameters.

$$P(X_1 = 0, X_2 = 0) = \pi_{00}$$

$$P(X_1 = 0, X_2 = 1) = \pi_{01}$$

$$P(X_1 = 1, X_2 = 0) = \pi_{10}$$

$$P(X_1 = 1, X_2 = 1) = \pi_{11} = 1 - \pi_{00} - \pi_{01} - \pi_{10}$$

W7-V1(c)

Use maximum likelihood to estimate all the parameters above.

W7-V1(c)

Use maximum likelihood to estimate all the parameters above.

$$\begin{aligned} p(\mathcal{D}|\theta) &= \prod_{i=1}^N p(x_1^{(i)}, x_2^{(i)}, y^{(i)}; \theta) \\ &= \prod_{i=1}^N p(y^{(i)}|x_1^{(i)}, x_2^{(i)})p(x_1^{(i)}, x_2^{(i)}) \\ &= \prod_{i=1}^N (\theta_{x_1^{(i)}, x_2^{(i)}})^{y^{(i)}} (1 - \theta_{x_1^{(i)}, x_2^{(i)}})^{(1-y^{(i)})} \pi_{x_1^{(i)}, x_2^{(i)}} \end{aligned}$$

Taking the logarithm, we have:

$$\ell(\theta) = \sum_{i=1}^N \left(y^{(i)} \log \theta_{x_1^{(i)}, x_2^{(i)}} + (1 - y^{(i)}) \log(1 - \theta_{x_1^{(i)}, x_2^{(i)}}) + \log \pi_{x_1^{(i)}, x_2^{(i)}} \right)$$

W7-V1(c)

Note that this objective decomposes, so we can estimate the parameters for each distribution separately

$$\begin{aligned}\ell(\pi_{00}, \pi_{01}, \pi_{10}, \pi_{11}) &= \sum_{i=1}^N \log \pi_{x_1^{(i)}, x_2^{(i)}} + C_0 \\ &= N_{00} \log \pi_{00} + N_{01} \log \pi_{01} + N_{10} \log \pi_{10} \\ &\quad + N_{11} \log(1 - \pi_{00} - \pi_{01} - \pi_{11}) + C_0\end{aligned}$$

Where N_{ij} is the number of data points with $X_1 = i$ and $X_2 = j$, and C_0 is independent of the π_{ij} s. Thus, to maximize ℓ with respect to the π_{ij} s, we take each

$$\frac{\partial \ell}{\partial \pi_{ij}} = 0$$

W7-V1(c)

Start with π_{00}

$$0 = \frac{\partial \ell}{\partial \pi_{00}} = \frac{N_{00}}{\pi_{00}} - \frac{N_{11}}{1 - \pi_{00} - \pi_{01} - \pi_{10}}$$

$$N_{00}(1 - \pi_{00} - \pi_{01} - \pi_{10}) = \pi_{00}N_{11}$$

$$N_{00}\pi_{11} = \pi_{00}N_{11}$$

Similarly for π_{01} and π_{10} .

So we have three equations and three unknowns! We can solve this!

W7-V1(c)

Since there is symmetry in the problem, we will only show the solution for π_{11} :

$$N_{00}\pi_{11} = \pi_{00}N_{11}$$

$$N_{01}\pi_{11} = \pi_{01}N_{11}$$

$$N_{10}\pi_{11} = \pi_{10}N_{11}$$

Add the three equations to get:

W7-V1(c)

Since there is symmetry in the problem, we will only show the solution for π_{11} :

$$N_{00}\pi_{11} = \pi_{00}N_{11}$$

$$N_{01}\pi_{11} = \pi_{01}N_{11}$$

$$N_{10}\pi_{11} = \pi_{10}N_{11}$$

Add the three equations to get:

$$(N_{00} + N_{01} + N_{10})\pi_{11} = (\pi_{00} + \pi_{01} + \pi_{10})N_{11}$$

$$(N_{00} + N_{01} + N_{10})\pi_{11} = (1 - \pi_{11})N_{11}$$

$$(N_{00} + N_{01} + N_{10} + N_{11})\pi_{11} = N_{11}$$

$$\pi_{11} = \frac{N_{11}}{N_{00} + N_{01} + N_{10} + N_{11}}$$

W7-V2(a,b)

How would you be able to compute $P(Y|X_1, X_2)$ in terms of $P(X_1, X_2|Y)$?
You should end up including $P(X_1, X_2)$ in your answer. How can you compute $P(X_1, X_2)$ in terms of the other quantities?

W7-V2(a,b)

How would you be able to compute $P(Y|X_1, X_2)$ in terms of $P(X_1, X_2|Y)$? You should end up including $P(X_1, X_2)$ in your answer. How can you compute $P(X_1, X_2)$ in terms of the other quantities?

Using Bayes Rule, $P(Y|X_1, X_2) = \frac{P(X_1, X_2|Y)P(Y)}{P(X_1, X_2)}$. Thus, we will need to be able to compute $P(Y)$.

Also, $P(X_1, X_2) = P(X_1, X_2|Y = 1)P(Y = 1) + P(X_1, X_2|Y = 0)P(Y = 0)$.

W7-V2(c)

What distribution does each of $P(X_1, X_2 | Y = 1)$ and $P(X_1, X_2 | Y = 0)$ have? And what parameters?

W7-V2(c)

What distribution does each of $P(X_1, X_2|Y = 1)$ and $P(X_1, X_2|Y = 0)$ have? And what parameters?

Categorical distribution, each with three parameters. Let's say:

$$P(X_1 = 0, X_2 = 0|Y = 1) = \theta_{00}$$

$$P(X_1 = 1, X_2 = 0|Y = 1) = \theta_{10}$$

$$P(X_1 = 0, X_2 = 1|Y = 1) = \theta_{01}$$

$$P(X_1 = 1, X_2 = 1|Y = 1) = \theta_{11} = 1 - \theta_{00} - \theta_{10} - \theta_{01}$$

and

$$P(X_1 = 0, X_2 = 0|Y = 0) = \phi_{00}$$

$$P(X_1 = 1, X_2 = 0|Y = 0) = \phi_{10}$$

$$P(X_1 = 0, X_2 = 1|Y = 0) = \phi_{01}$$

$$P(X_1 = 1, X_2 = 1|Y = 0) = \phi_{11} = 1 - \phi_{00} - \phi_{10} - \phi_{01}$$

W7-V2(d)

How many parameters in total need to be learned? How does this compare with the number of parameters learned in the previous question?

W7-V2(d)

How many parameters in total need to be learned? How does this compare with the number of parameters learned in the previous question?

There should be 7 parameters (because $P(Y)$ requires one more).

This is the same as the number of parameters in the previous question.

W7-V2(e)

Use maximum likelihood to estimate all the parameters above.

W7-V2(e)

Use maximum likelihood to estimate all the parameters above.

$$\begin{aligned} p(\mathcal{D}|\theta) &= \prod_{i=1}^N p(x_1^{(i)}, x_2^{(i)}, y^{(i)}; \theta) \\ &= \prod_{i=1}^N p(x_1^{(i)}, x_2^{(i)} | y^{(i)}) p(y^{(i)}) \\ &= \prod_{i=1}^N (\theta_{x_1^{(i)}, x_2^{(i)}})^{y^{(i)}} (\phi_{x_1^{(i)}, x_2^{(i)}})^{(1-y^{(i)})} \pi^{y^{(i)}} (1 - \pi)^{(1-y^{(i)})} \end{aligned}$$

... Taking the logarithm, we have:

$$\begin{aligned} \ell(\theta) &= \sum_{i=1}^N (y^{(i)} \log \theta_{x_1^{(i)}, x_2^{(i)}} + (1 - y^{(i)}) \log(1 - \phi_{x_1^{(i)}, x_2^{(i)}}) + \\ &\quad y^{(i)} \log \pi + (1 - y^{(i)}) \log(1 - \pi)) \end{aligned}$$

Again, we can estimate the parameters for each distribution separately.

During the lecture and the Naive Bayes lab, we classify whether movie reviews were positive or negative using the Naive Bayes model:

$$p(c, x_1, \dots, x_D) = p(c)p(x_1|c) \cdots p(x_D|c)$$

Where c is the label class (positive or negative), and each $x_j \in \{0, 1\}$ represents whether word j appears in the review. We will fit parameters π and θ_{jc} , with $p(c = 1) = \pi$, representing the probability of positive class and $p(x_j = 1|c) = \theta_{jc}$, representing the probability of word j appearing in a review with class c .

Suppose that we would like to find the word whose **presence** most strongly indicate that the review is positive. In other words, we wish to find the word j with the largest value of $p(c = 1|x_j = 1)$.

Express the quantity $p(c = 1|x_j = 1)$ in terms of the fit parameters θ_{jc} and π .

Express the quantity $P(c = 1|x_j = 1)$ in terms of the fit parameters θ_{jc} and π by Using Bayes' Rule:

$$P(c = 1|x_j = 1) = \frac{P(x_j = 1|c = 1)P(c = 1)}{P(x_j = 1)}$$

where $P(x_j = 1)$ is:

$$P(x_j = 1) = P(x_j = 1|c = 1)P(c = 1) + P(x_j = 1|c = 0)P(c = 0)$$

Putting this back into Bayes' Rule

$$P(c = 1|x_j = 1) = \frac{P(x_j = 1|c = 1)P(c = 1)}{P(x_j = 1|c = 1)P(c = 1) + P(x_j = 1|c = 0)P(c = 0)}$$

Putting this back into Bayes' Rule

$$P(c = 1|x_j = 1) = \frac{P(x_j = 1|c = 1)P(c = 1)}{P(x_j = 1|c = 1)P(c = 1) + P(x_j = 1|c = 0)P(c = 0)}$$

Having $P(c = 1) = \pi$, $P(x_j = 1|c = 1) = \theta_{j1}$, $P(c = 0) = 1 - \pi$ and $P(x_j = 1|c = 0) = \theta_{j0}$ the equation is:

$$P(c = 1|x_j = 1) = \frac{\theta_{j1}\pi}{\theta_{j1}\pi + \theta_{j0}(1 - \pi)}$$